

Cylinder Experiment Report Guidelines

Experiment Objectives

Recall that the objectives of this experiment were to demonstrate characteristics of viscous fluid flow around a circular cylinder (e.g., pressure profiles and boundary layer flows) and to evaluate the drag force exerted on a cylinder.

Presentation of Results

Your results are to be presented to your TA as a written memo report. Guidelines for this report are provided separately. This report is to be submitted as a pdf file to Canvas by 11:59 pm on May 18, 2026. The late policy for this report is 25% off (of the total possible) for every 24-hour period that the report is late. As an example, if you turn in your report after 11:59 pm on May 18, 2026, but by 11:59 pm on May 19, 2026, the 25% penalty will be applied.

Data Reduction and Analysis

1. Plot the measured gage pressures (in Pa) as a function of the angle θ on the cylinder surface.
2. Based on your experimental data, plot the pressure coefficient C_p as a function of the angle θ . Show on the same plot the inviscid flow C_p values as a function of θ . Compare the experimental C_p variation with θ with that predicted by inviscid flow theory. Explain differences that may exist.
3. Using your experimental data on surface pressure as a function of angle, compute the pressure drag force D_p acting on the cylinder. Do this by

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numerically integrating the surface pressure data over the cylinder surface.

4. Using published data on drag coefficients for flow over cylinders, calculate a theoretical value for the total drag force (D_t) acting on the cylinder.
5. Compare and contrast the theoretical drag force, the pressure drag force, and the measured drag force. Discuss reasons for differences.
6. From your pressure and flow visualization data, estimate the angle at which flow separation occurred. Does your estimate agree with published data? Provide citations. If significant differences exist, discuss possible reasons for these differences.
7. From your data on the effect of pitot-static probe angle (from horizontal) on the manometer reading, calculate the apparent air speed at each of the angles studied. Discuss the effects of misalignment of the pitot-static probe.
8. From your data on the effect of pitot-static tube lateral traverse on the manometer readings, evaluate the air speed at the different lateral positions studied. Discuss whether the velocity profile was uniform upstream of the cylinder.

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Equations

Experimental Pressure Coefficient	$C_p = (p_s - p_\infty) (\rho V^2/2)^{-1}$
Inviscid-Flow Pressure Coefficient	$C_p = 1 - 4 \sin^2\theta$
Pressure Drag	$D_p = 2RL \int_0^\pi (p_s - p_\infty) \cos(\theta) d\theta$

Nomenclature

L = cylinder length = 300 mm

R = cylinder radius = 31.8 mm

p_s = local surface pressure

p_∞ = static pressure far upstream of the cylinder

V = speed of the flow far upstream of the cylinder

θ = angle from the forward stagnation point

ρ = air density